

An Investigation of Harmonic Oscillators

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1 Introduction

The goal of this experiment is to investigate the fundamental properties of harmonic oscillators; their spring constants k , damping factors B and their motion in series and parallel. (k), is a measure of a springs stiffness, indicating how much force is needed to compress or stretch the spring by a unit distance and is measured in Nm^{-1} . The factors that determine the spring constant are:

- The Material's Properties: The level of stiffness is primarily dictated by the material's intrinsic characteristics, such as its Young's modulus.[2]
- The Spring's Dimensions: The numerical value of k is also altered by the spring's discipline of geometry, encompassing its length, coil diameter, and dimension. [2]

Motivation:

Oscillators play a vital role in physics because their restoring forces allow complex systems to be modelled using simple harmonic motion, such as inter-atomic forces or practical applications like shock absorbers for buildings [4]. For these reasons it is useful to be able to understand and model their behaviour in a simple way. These experiments aim to investigate and confirm some of the known laws and equations for systems of harmonic oscillators.

2 Theory and Experiment

2.1 Part 1: Determination of the spring constant using Hooke's law

2.1.1 Theory

From Hooke's Law the restoring force, F_x is given by:

$$F_x = -kx$$

Plotting a graph of F_x vs the extension of the spring x should yield linear trend through the origin with a slope equal to k .

2.1.2 Experiment

The experiment was carried out as described in Part 1 of the lab manual [1]. A metre stick was used to evaluate the extension of the spring.

2.2 Part 2: Determination of the spring constant from the frequency of oscillation

2.2.1 Theory

From the differential equation of simple harmonic motion the frequency of oscillation, ω_0 is given by:

$$\omega_0^2 = \frac{k}{m}$$

ω_0 can then be multiplied by the mass m attached to the oscillator to determine k .

2.2.2 Experiment

The experiment was carried out as described in Part 2 of the lab manual [1].

2.3 Part 3: Determination of the damping factor of a system consisting of one mass and one oscillator

2.3.1 Theory

Due to air resistance and the internal friction of the oscillator, the motion will be damped and the amplitude of the oscillations will decrease exponentially according to the equation;

$$x(t) = Ae^{Bt} \sin(\omega_1 t + \Phi)$$

Where B is the damping factor and ω_1 is the damped angular frequency.

2.3.2 Experiment

A mass of $50g$ was suspended from the spring and the oscillator was extended such that it did not exceed its elastic limit. The system was left to oscillate for a period of $100s$ and the amplitude was logged 10 times a second using the magnetic field sensor. A damped sinusoidal curve was fitted to the data and the damping coefficient B was found using the equation of this curve. This process was repeated 5 times and the results averaged to determine B of the system.

2.4 Part 4: Determination of the spring constant of a system consisting of one mass and two oscillators

2.4.1 Theory

In a system consisting of two identical oscillators with spring constant k the effective spring constant is defined as follows; In series;

$$\frac{1}{k_{eff}} = \frac{1}{2k}$$

In parallel;

$$k_{eff} = 2k$$

2.4.2 Experiment

The experiment was carried out as described in Part 4 of the lab manual [1].

2.5 Part 5: Investigation into the motion of a system consisting of two masses and two oscillators

2.5.1 Theory

When two masses are connected by oscillators, the system no longer behaves as a single simple harmonic oscillator. Instead, the motion is a superposition of two normal modes, each with its own characteristic angular frequency. These normal modes arise because the displacement of one mass influences the motion of the other through their coupling. The equations describing these modes are available in Part 5 of the lab manual [1].

2.5.2 Experiment

The experiment was carried out as described in Part 5 of the lab manual [1].

3 Results and Analysis

3.1 Part 1: Determination of the spring constant using Hooke's law

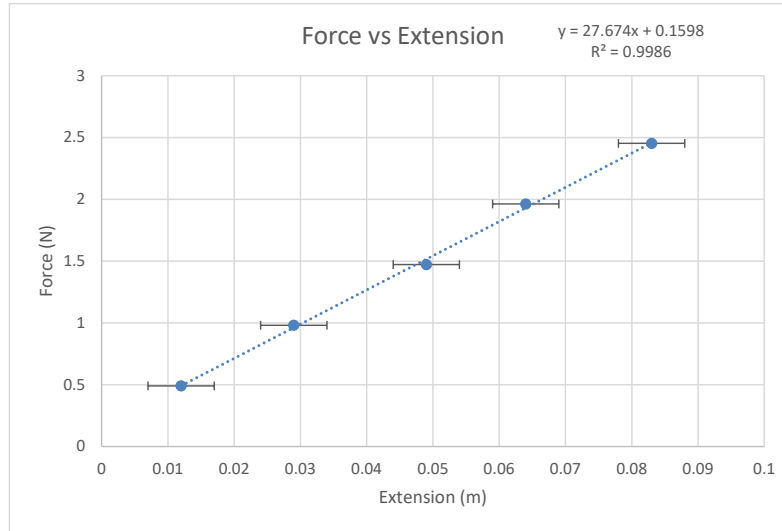


Figure 1: Graph of Force vs Extension

The force and respective extension was plotted and the linear trend was found to be;

$$y = 27.674x + 0.1598$$

This implies that $k = 27.674 \text{ N/m}$, given that k is found from this linear trend the error in k can be taken to be the error in the slope which was found to be;

$$\sigma_k = 0.592 \text{ Nm}^{-1}$$

This results in a spring constant of;

$$k = 27.67 \pm 0.59 \text{ Nm}^{-1}$$

3.2 Part 2: Determination of the spring constant from the frequency of oscillation

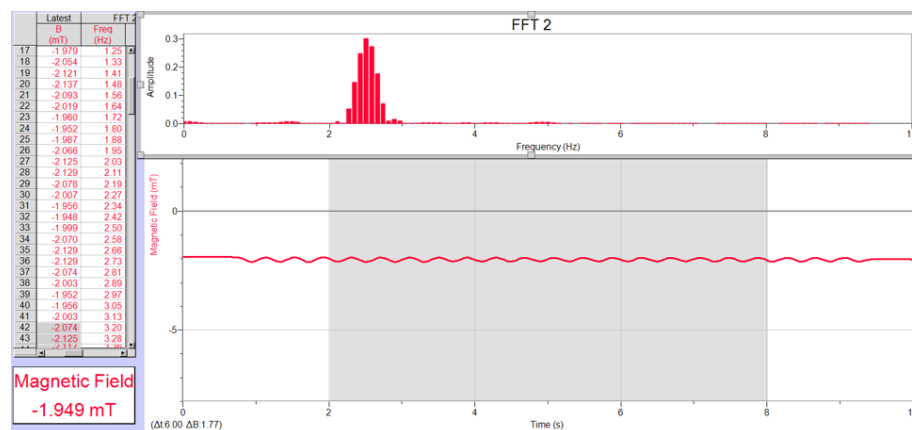


Figure 2: FFT Graph and Graph of B vs t

Part 2: Frequency Method – Dynamic Determination of k				
Mass m (kg)	Frequency f (Hz)	$\omega = 2\pi f$ (rad/s)	ω^2 (rad ² /s ²)	k = m ω^2 (N/m)
0.05	3.6154	22.71622816	516.0270218	25.80135109
0.1	2.5285	15.88703405	252.3978509	25.23978509
0.15	2.1538	13.53272451	183.1346328	27.47019492

Mean k (N/m)	26.1704437
Std dev k	1.160109485
Standard error	0.669789524
k $\pm$ SE (N/m)	26.17 $\pm$ 0.67

Comparison w Part 1	Good agreement (<10% diff)
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Figure 3: Results of Part 2

The average of the trials was taken resulting in;

$$k = 26.170 \text{ Nm}^{-1}$$

The standard error of the trials was taken to be the error in k resulting in a final value of;

$$k = 26.17 \pm 0.67 \text{ Nm}^{-1}$$

This is comparable to the spring constant found in part 1 and the discrepancy will be discussed further in the discussion section.

3.3 Part 3: Determination of the damping factor of a system consisting of one mass and one spring

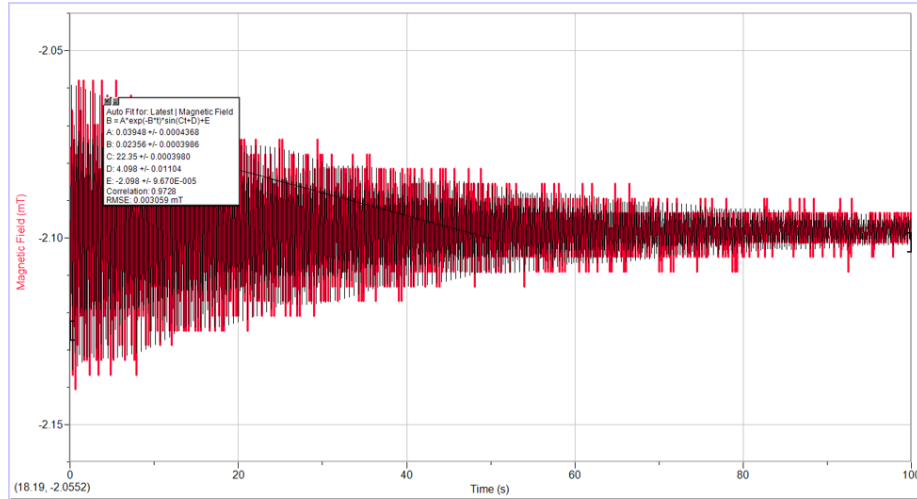


Figure 4: Graph of Magnetic Field vs Time with Fitted Damped Sinusoidal Curve

The damping factor was found to be;

$$\beta = 0.02154 \text{ s}^{-1}$$

The error in the coefficient that provided β was given by the software when the curve was fitted and so the final value of the damping factor was found to be;

$$\beta = 0.022 \pm 0.001 \text{ s}^{-1}$$

3.4 Part 4: Determination of the spring constant of a system consisting of one mass and two oscillators

Part 4: Springs in Series and Parallel				
Configuration	Mass m (kg)	Frequency f (Hz)	$\omega = 2\pi f$ (rad/s)	$k_{\text{eff}} = m\omega^2$ (N/m)
Single spring	0.05	3.5385	22.23305121	24.7154283
Series	0.05	2.4615	15.46606063	11.95995158
Parallel	0.1	3.61538	22.7161025	51.60213126
Expected k_{series} (N/m)	12.35771415			
Expected k_{parallel} (N/m)	49.43085661			
Measured k_{series} (N/m)	11.95995158			
Measured k_{parallel} (N/m)	51.60213126			
% diff series	3.218739089			
% diff parallel	4.39254911			

Figure 5: Results of Part 4

3.5 Part 5: Investigation into the motion of a system consisting of two masses and two oscillators

Part 5: Coupled Oscillators					
Baseline:					
Mass m (kg)	0.05				
Single-spring k (N/m)	26.92				
$\omega_0^2 = k/m$ (rad ² /s ²)	538.4432487				
$\omega_1^2 = (3+\sqrt{5})/2 \cdot \omega_0^2$	1409.662726				
$\omega_2^2 = (3-\sqrt{5})/2 \cdot \omega_0^2$	205.66702				
$f_1 = \omega_1/2\pi$ (Hz)	5.975548007				
$f_2 = \omega_2/2\pi$ (Hz)	2.282456237				
Equal masses – Mode 1	Average Measured f (Hz)	$\omega = 2\pi f$	ω^2	Theoretical ω^2	% diff
	2.2143	13.91285723	193.5675962	205.66702	5.88
Equal masses – Mode 2	Average Measured f (Hz)	$\omega = 2\pi f$	ω^2	Theoretical ω^2	% diff
	5.8333	36.65170485	1343.347469	1409.662726	4.70
Unequal masses – Heavy on top, Mode 1	Average Measured f (Hz)	$\omega = 2\pi f$			
	1.9286	12.11775118			
Unequal masses – Heavy on top, Mode 2	Average Measured f (Hz)	$\omega = 2\pi f$			
	4.6905	29.47128068			
Unequal masses – Light on top, Mode 1	Average Measured f (Hz)	$\omega = 2\pi f$			
	1.7143	10.77126457			
Unequal masses – Light on top, Mode 2	Average Measured f (Hz)	$\omega = 2\pi f$			
	5.4286	34.10889976			
Same result when reversed?	Yes (within 20%)				

Figure 6: Results of Part 5

4 Discussion and Conclusion

4.1 Part 1: Determination of the spring constant using Hooke's law

The fitted line gave $R^2 = 0.9986$ indicating high precision and small random scatter in the measurements. However the intercept being at 0.1598 indicates some systematic offset. This could be due to the approximation of g to 9.81 ms^{-2} or consistent inaccuracies in the measurement of the extension. Another possible source of systematic error was the hysteresis effect. Hysteresis results from the internal friction that arises when the oscillator's material deforms and recovers [3]. For this experiment it would have been possible to limit the effects of hysteresis by first measuring the extension while loading the oscillator and then again while unloading it and averaging those results.

4.2 Part 2: Determination of the spring constant from the frequency of oscillation

The static and dynamic methods result in k that differs by about 6%. This is consistent with the assumptions made in part 2 to ignore the effective mass m_{eff} of the system and instead only include the hanging mass in the calculation. By doing some calculations we can approximate the effective mass of the system and compare that to the difference in the results from the two methods we can determine if the difference between the two methods is acceptable.

$$k_{dynamic} = m\omega^2 \quad (1)$$

$$k_{static} = m_{eff}\omega^2 \quad (2)$$

$$m_{eff} = m \frac{k_{static}}{k_{dynamic}} \quad (3)$$

Now substitute the experimental values into (3);

$$m_{eff} = 1.0574m$$

So the effective mass of the system is approximately 5.7% heavier than the hanging mass alone, from this calculation the 6% difference between the two methods seems acceptable. For a repeat experiment the effective mass would be incorporated into the calculations

4.3 Part 3: Determination of the damping factor of a system consisting of one mass and one oscillator

The accuracy of this part of the experiment was quite high approximately 1000 data points were used to calculate the damping coefficient in each trial. This means that the standard error of the mean is accurate to around 2%. Which is significantly smaller than the systematic uncertainties such as the FFT resolution. The precision was slightly lower than expected but upon reflection this

should have been expected. Although the damping factor is a property of the system and does not depend on the initial extension, the measured damping factor can vary with initial displacement. (Large amplitudes introduce non-linear spring behaviour and horizontal motion, while small amplitudes reduce the signal-to-noise ratio.) These effects can influence the fitted curve and therefore the calculated damping factor. To increase precision it would be wise to have the same initial extension on every trial of the experiment.

4.4 Part 4: Determination of the spring constant of a system consisting of one mass and two oscillators

The measured values the oscillators in both series and parallel were within 5% of the theoretical values, as the precision was high (the same frequencies were observed in every trial) the most likely sources of this uncertainty are in the resolution of the FFT graph or as discussed in part 2 the effective mass on the system.

4.5 Part 5: Investigation into the motion of a system consisting of two masses and two oscillators

Equal Masses:

For the coupled oscillators with equal masses the measured frequencies were within 6% of the theoretical values, similarly to part 4 the precision was high and the most likely sources of experimental error were the same. However these theoretical values were calculated using the spring constant found in part 2 which as already discussed is most likely about 6% inaccurate meaning the comparison could be more accurate.

Un-Equal Masses:

The data shows that reversing the masses does not change the normal-mode frequencies, since the theoretical expressions are symmetric in m_1 and m_2 . However the observed motion and the FFT peak amplitudes differ between configurations because the mode shapes change the lighter mass oscillates with a larger amplitude. The precision was once again found to be high and causes of uncertainty were most likely those discussed in part 4.

5 Bibliography

Whole bibliography

- [1] UCC Physics Department. “py2107 experiment2 v3”. In: *PY2107 Canvas Page* (2026).
- [2] Rabina Kadariya. *Hooke’s Law: Development, Applications and Limitations*. URL: <https://scienceinfo.com/hookes-law/>. (accessed: 21.09.2026).
- [3] *Minimizing Hysteresis in Technical Springs*. URL: <https://www.tevema.com/minimizing-hysteresis-technical-springs/#:~:text=Hysteresis%20results%20from%20the%20internal%20friction%20that%20arises,also%20lead%20to%20fatigue%20and%20failure%20over%20time..> (accessed: 24/09/2026).
- [4] *Real Life Applications of Hooke’s Law*. URL: <https://www.geeksforgeeks.org/physics/real-life-applications-of-hookes-law/>. (accessed: 21.09.2026).